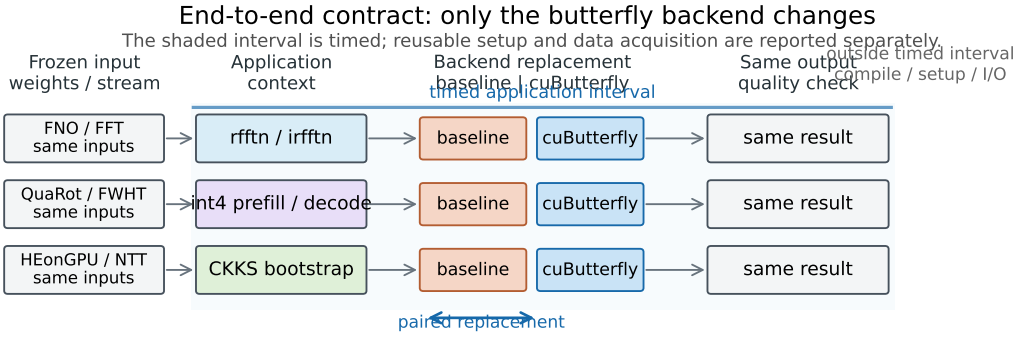
End-to-end contract: only the butterfly backend changes
The shaded interval is timed; reusable setup and data acquisition are reported separately.
Frozen input weights / stream
Application context
Backend replacement baseline | cuButterfly
Same output quality check
outside timed interval
compile / setup / I/O
timed application interval
FNO / FFT same inputs
rfftn / irfftn
baseline
cuButterfly
same result
QuaRot / FWHT same inputs
int4 prefill / decode
baseline
cuButterfly
same result
HEonGPU / NTT same inputs
CKKS bootstrap
baseline
cuButterfly
same result
paired replacement